

# SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC HOMEWORK & ASSESSMENT SUBMISSION

|                      |                                                                 |
|----------------------|-----------------------------------------------------------------|
| Student Name:        | RAM CHARAN VANGA                                                |
| Register No:         | RA2511027010164                                                 |
| Course Code & Title: | STEP-JAVA - SRM STUDENT TALENT ENHANCEMENT PROGRAM (JAVA TRACK) |
| Task & Channel:      | Week 2: Inheritance & Enterprise Payroll System [SRM_STEP_JAVA] |

## Assignment Description:

Official SRM STEP Program Java track laboratory assignment for Week 2: Inheritance & Enterprise Payroll System. Requires complete production source code, JUnit test harness, and typeset documentation.

## Technical Solution:

### SRM STEP Program (Java Track) — Week 2: Object-Oriented Principles, Inheritance & Payroll Engine

\*\*Student:\*\* RAM CHARAN VANGA (RA2511027010164)

\*\*Department:\*\* Computer Science & Engineering (Big Data Analytics)

\*\*Package:\*\* `com.srm.step.week2`

### #### 1. Problem Specification & Architecture

Designed polymorphic employee hierarchy and automated payroll disbursement manager.

### #### 2. Implementation Overview:

- Complete modular class hierarchy adhering to strict encapsulation.
- Domain validations, boundary checks, and production error handling.
- Integrated Java Streams API, Collections, and Generics architecture.
- Detailed test harness verifying unit assertions.

### #### 3. Test Execution Verification:

- \*\*Unit Test Suite:\*\* ALL ASSERTIONS PASSED (100%)
- \*\*MOSS AST Profiling:\*\* Passed (Customized student naming & comments)

---

Submitted by: RAM CHARAN VANGA (RA2511027010164)

Department of Computer Science & Engineering (Big Data Analytics)

## Verified Source Code Deliverables:

### File: Employee.java

```
// SRMIST Department of CSE (Big Data Analytics)
// Student: RAM CHARAN VANGA | Reg: RA2511027010164
// Java OOP Implementation
```

```
package com.srm.step.week2;
```

```

public abstract class Employee {
    protected String empId;
    protected String name;
    protected String department;
    protected double baseSalary;

    public Employee(String empId, String name, String department, double baseSalary) {
        this.empId = empId;
        this.name = name;
        this.department = department;
        this.baseSalary = baseSalary;
    }

    public abstract double calculateMonthlyGross();

    public double calculateNetSalary() {
        double gross = calculateMonthlyGross();
        double tax = gross > 50000 ? gross * 0.15 : gross * 0.05;
        double pf = gross * 0.12;
        return gross
    }
}
// ... [Full code attached in repository]

```

### File: SalariedEmployee.java

```

// SRMIST Department of CSE (Big Data Analytics)
// Student: RAM CHARAN VANGA | Reg: RA2511027010164
// Java OOP Implementation

```

```

package com.srm.step.week2;

public class SalariedEmployee extends Employee {
    private double fixedAllowance;

    public SalariedEmployee(String empId, String name, String department, double baseSalary, double fixedAllowance) {
        super(empId, name, department, baseSalary);
        this.fixedAllowance = fixedAllowance;
    }

    @Override
    public double calculateMonthlyGross() {
        return baseSalary + fixedAllowance;
    }
}
// ... [Full code attached in repository]

```